Group 10: Zoe Garbus, John Kraunelis, Isha Venkatapathy

Professor Xiaoyi Yang

DS4200: Information Presentation and Visualization

13 April 2025

A Data-Driven Analysis of Supply Chain Management Practices

Link to published website: https://jokraun12.github.io/ds4200-final-project/

I. Box Plot: Comparing Operational Efficiency by SCM Practice

The box plot is a clear visual comparison on how operational efficiency scores vary amongst the different SCM practices. Our dataset measured operational efficiency as the ratio of outputs to inputs which is calculated by comparing a company's revenues to its operating expenses and costs of goods sold. A score of 100 means perfect efficiency which is the company maximizing its operational productivity with minimal waste and cost. This is a pretty useful metric to have in evaluating supply chain performance since it captures the main goal of maximizing value while minimizing costs. We chose a box plot for this analysis because it effectively shows not just the central tendencies but also the spread, quartiles, and potential outliers in an easily comparable format.

Most of the SCM practices have a median efficiency score of ~85 which suggests a general standard around this value. The most notable part of this visual is the variation in distribution shapes across each of the practices, While the medians scores for each SCM do cluster around 85, the distribution widths tell a more nuanced story on the performance reliability Agile SCM shows the most variability ranging from 75 to 90 operational efficiency score. This wider spread means that the implementation quality of the practice plays a big role in its outcome. This variability aligns with Agile's emphasis on flexibility where different implementations do lead to significantly different results based on the execution specifics. Cross-docking has the narrowest distribution from 80-85 which suggests that it is much more consistent in operational outcomes regardless of the company that is implementing the practice. Compared to Agile, Cross-Docking likely delivers much more predictable efficiency results with less sensitivity to the implementation variables.

It is important to note that all of the practices achieve maximum efficiency scores of around 88-90 which shows that operational excellence is possible regardless of the approach being used. The variability we see in the distribution for each SCM does indicate that some practices have more reliable pathways to efficiency while others have a larger range depending on the quality of the implementation. The interactive selection feature further enhances the analytical capabilities of this chart by allowing users to compare specific practices directly or view them all together. This is useful in making a more direct comparison between practices by spotting subtle differences between the selected SCMs.

II. Bubble Chart: SCM Strategies of Market Leaders

The primary goal of this visualization was to illustrate the distribution and adoption of various SCM practices among leading U.S. corporations. It highlights how different SCM strategies are chosen based on industry needs, company size, and the amount of suppliers. A bubble chart was useful for identifying these trends because it can visualize multiple dimensions simultaneously. It shows the size of the market cap through the size of the bubble, the color to indicate the SCM practice the company uses, and the tooltip shows the supplier count. Distinct coloring of the bubbles assigned to each SCM practice allows more immediate and apparent analysis of the use of each, with a clear label to identify the company.

The motivation behind this design was to make visible the strategic choices that major companies make in managing their supply chains and to examine whether some approaches are more dominant due to their effectiveness in handling complex operations. The visualization also serves to spotlight any outliers, such as Google's unique use of Cross-Docking alongside an exceptionally large supplier base (~20,000). These types of details help emphasize how SCM practices are often tailored to specific business models, market conditions, and logistical needs.

From the visualization we can tell that the most prominent SCM strategies appear to be Agile SCM and Lean Manufacturing. The less common practices are Vendor Managed Inventory and Efficient Consumer Response. This suggests that certain SCM strategies are more widely adopted by industry leaders, potentially due to their scalability and efficiency in handling global supply chains. Companies using Agile SCM (Apple, Amazon, Tesla) prioritize rapid response to demand changes, customization, and flexibility in their supply chains. This more flexible strategy is particularly suited for industries where innovation cycles are fast and consumer demand fluctuates. Lean Manufacturing (Nvidia, Microsoft, Facebook) focuses on minimizing waste, reducing inefficiencies, and creating value for customers to enhance cost efficiency and maintain a steady output. It is valuable for industries where demand is relatively predictable and manufacturing processes can be streamlined.

III. Scatter Plot: Comparing Customer Satisfaction and Inventory Turnover

Inventory Turnover Ratio can be defined as the Average Cost of Goods Sold divided by the Average Inventory on Hand. This ratio is a practical way to get a sense of how much inventory a firm is turning over on average. This is especially important from a Supply Chain Management standpoint, as this discipline has to do with the distribution systems within a firm, specifically how efficient they are at receiving inventory and then shipping it out to customers.

However, as we speak of on the website, the inventory turnover ratio can be a slightly misleading number at times. While generally speaking, a higher ratio indicates a more efficient SCM process, sometimes a ratio can be diluted by firms that do not necessarily have enough inventory on hand to meet consumer demands.

The thought process behind the scatter plot design revolved around the goal of visualizing each supply chain management practice from an objective standpoint. From a proprietary financial standpoint, the inventory turnover ratio is a particularly objective value to analyze. Secondly, a customer satisfaction rating is a second independent variable that is relatively objective for each SCM practice, as well as easy to understand for viewers. Therefore, the goal here was to show the positive trend between higher inventory turnover ratios and higher customer satisfaction rating. In doing so, we were also able to identify an example of the previously stated exception. While the positive trend was correctly identified, we also were able to visualize the "Efficient Consumer Response" SCM practice as an outlier. This SCM practice had an inventory turnover ratio of 10.025, the highest out of all SCM categories, however it had the lowest average customer satisfaction rating. Efficient Consumer Response has to do with managing a firm's supply chain as dependent on consumer needs. It is likely that many firms that utilize this practice struggle with timing their inventory inflows correctly with their consumer demands.

IV. Line Plots: Relationship between SCM Practice & Financial Performance

Our goal of incorporating financial data into our SCM dataset was to visualize more key metrics, specifically timestamped data so we could identify familiar trends amongst different SCM practices. To do this, as mentioned on our website, we utilized HTTP GET requests to Yahoo Finance's API to get stock tickers for each company name in our dataset. Then, utilizing the Python library *yfinance,* we were able to get observations on which companies were listed on US stock exchanges. Out of the 421 firms listed on US stock exchanges, we identified the top company in each SCM category by highest operational efficiency score. For these firms, we downloaded share price data utilizing the same Python library from 2018 to 2022. The goal of using this timeline was to get an accurate visual on the performance of these firms during the COVID-19 pandemic, a time when the global supply chain was upended. To compare these firms to a benchmark, we plotted out the share price of State Street's SPY ETF, a fund that follows the performance of the S&P500, which contains the shares of the top 500 largest companies on the NYSE.

The first line plot illustrates the share price overtime for these firms, while the second depicts the *cumulative return* on these firms from inception until the end of 2022. This is a way that we normalized our data, as different firms may have wildly different share price ranges, but cumulative return offers a way for us to see an objective performance of the company stock overtime. This is derived by finding the cumulative product of the percent change in share price over each day, for each company. Then, we decided to zoom in on the March 2022 market drop caused by after-effects of the COVID-19 pandemic. Then from here, we were able to analyze the length of recovery time for each best performer from each SCM category. In theory, our design for this visualization revolved around comparing the length of recovery time for each SCM practice to the time it took for the market to recover. We theorized that more efficient and resilient SCM practices would have recovered faster than the benchmark, whereas less efficient and resilient SCM practices would have underperformed the market in terms of recovery time.